

Mapudungun–Spanish Speech Translation: A Low-Resource End-to-End System for the IWSLT 2026 Shared Task

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Abstract

We present an end-to-end speech translation system for Mapudungun–Spanish developed for the IWSLT 2026 low-resource task. Building on the Canary-1B-v2 model, we apply parameter-efficient fine-tuning with a lightweight adapter and leverage an English-centered configuration as a proxy to enable translation. Experiments show that the system captures key phonetic patterns despite limited data, though it exhibits biases toward repetitive Spanish outputs. Our results highlight both the feasibility and the challenges of adapting multilingual foundation models to low-resource Indigenous languages.¹

1 Introduction

This work summarizes our entry for the Low-Resource Speech Translation shared task at the International Conference on Spoken Language Translation (IWSLT) 2026. This shared task aims to benchmark and advance speech translation for diverse dialects and low-resource languages, where limited parallel data constrain standard supervised approaches. It emphasizes the need for novel methods that leverage heterogeneous resources to address data scarcity. The task comprises two tracks: (1) a traditional speech-to-text translation track, and (2) a data track focused on the creation and open release of speech translation datasets for under-resourced languages. The current edition explicitly prioritizes multilingual systems capable of handling a broad range of languages, encouraging participants to develop generalizable approaches and engage with multiple language pairs, while still allowing specialized systems for individual pairs.

Our entry focuses on Task 1 and on a single language pair. We build an end-to-end speech trans-

lation system from Mapudungun (arn) to Spanish (spa) using the Canary-1B-v2 model.

In the next section 2, we describe the main characteristics of the Mapudungun language and corpus. Section 3 presents our approach, Section 4 reports our results, Section 5 provides our concluding remarks, and Section 6 discusses the limitations of our study.

2 Mapudungun language and corpus

Mapudungun (arn) is an Indigenous language of the Americas spoken by the Mapuche people in south-central Chile and west-central Argentina. It has an estimated native speaker population of around 250,000. Under the Agglomerated Endangerment Scale (AES), Mapudungun is classified as *shifting (downward)*, meaning that the language is undergoing a process of increasing endangerment (Hammarström et al., 2024).

Our entry uses the provided Mapudungun corpus (Duan et al., 2020). The corpus consists of 142 hours of spoken Mapudungun, collected between 2001 and 2005 as part of the AVENUE project (Levin et al., 2000), in collaboration among Carnegie Mellon University, the Chilean Ministry of Education, and the Universidad de La Frontera. The data, which originally spanned 170 hours, were cleaned and publicly released with transcriptions and Spanish translations. The corpus covers three closely related dialects—Nguluche (approximately 110 hours), Lafkenche (approximately 20 hours), and Pewenche (approximately 10 hours).

The recordings are domain-specific, focusing on primary and traditional healthcare practices, and were collected through ethnographically grounded, open-ended conversational interviews. The dataset reflects culturally authentic interactions among native speakers, with participants aged 16 to 100 and balanced by gender. Ethical considerations were addressed through informed consent, anonymization,

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¹<https://github.com/umoqnier/iwstl-low-resource-experiments>

and the removal of sensitive cultural knowledge.

From a linguistic perspective, the corpus uses a non-standard, supra-dialectal orthography developed at the time of collection, consisting of 28 graphemes designed to be compatible with Spanish keyboards. Although this differs from the modern Azumcheffe standard, both the original and normalized versions are valuable. The dataset also includes rich annotations capturing disfluencies, non-verbal sounds, and code-switching (e.g., Spanish borrowings), making it a comprehensive and challenging resource for speech and language processing in a low-resource setting.

A substantial cleaning and normalization process was applied to make the corpus suitable for modern computational use. This includes fixing inconsistencies in utterance boundaries and speaker labels, resolving missing or mismatched files, converting text encodings to Unicode, standardizing formatting, and removing duplicates. Audio was normalized in power, and turn boundaries were adjusted to control leading and trailing silence. Additionally, forced alignment techniques were applied to improve synchronization between audio and transcripts, enabling the identification of well- and poorly aligned segments. Residual alignment issues are partly attributed to orthographic variation, which remains an area for future normalization.

3 Approach

Our approach focuses on fine-tuning NVIDIA’s Canary-1B-v2 model for speech-to-text translation (ST) (Sekoyan et al., 2025), a multilingual/multitask model trained for various tasks, including automatic speech recognition (ASR). Its implementation is based on the Conformer architecture (Gulati et al., 2020); specifically, its more efficient variant known as FastConformer (Rekesh et al., 2023). The original model released by NVIDIA (NVIDIA, 2025) was trained on a 1.7-million-hour dataset covering 22 high-resource and 3 low-resource languages. It uses both pseudo-labeled and human-annotated data and is also exposed to non-speech audio associated with empty outputs to improve its robustness.

The Canary model supports speech translation through English-centered language pairs. In one direction, the source language is English and the target can be any of 24 languages ($eng \rightarrow X$). In the second configuration, the direction is reversed,

from any of the 24 languages to English ($X \rightarrow eng$). As we show below, this design becomes a challenge because English is not part of our chosen language pair (arn, esp) for the shared task.

We chose the Canary model based on the following advantages:

- **Accessible codebase:** Canary uses the NeMo NVIDIA framework (Kuchaiev et al., 2019), which makes the training code readily accessible.
- **Spanish coverage:** Spanish is one of the languages covered by the model, so the system is already capable of "listening" and "generating" Spanish.
- **Hardware requirements:** The implementation is efficient, and it supports parameter-efficient fine-tuning (PEFT) that significantly reduces the amount of computer power; in our case, we perform experiments with two GPUS with 24GB of VRAM (12GB on each GPU) and a single GPU with 32GB of VRAM.
- **Preliminary testing:** Our initial experiments allowed us to easily prototype the scripts needed to train and evaluate the system.

3.1 Data preprocessing

The speech signal was augmented using an on-the-fly waveform augmentation pipeline (Jordal, 2023) applied directly during training, eliminating the need to store additional preprocessed data. Prior to batch collation and zero-padding, the raw audio signals were stochastically augmented through a combination of background noise injection, Gaussian noise, aliasing, room impulse responses, band-pass filtering, and pitch shifting. Additionally, in one experimental run, we applied a duration-based filtering step that discarded recordings longer than 15 seconds. The latter was done to reduce instabilities during training that would otherwise halt the process and to maintain more uniform sequence lengths within batches.

The Spanish translations were also preprocessed by eliminating capitalization and removing punctuation.

3.2 Architecture

The NeMo framework allows us to fine-tune the Canary-1B-v2 model with considerable flexibility. In particular, we adopt a parameter-efficient fine-tuning (PEFT) strategy (Houlsby et al., 2019) based

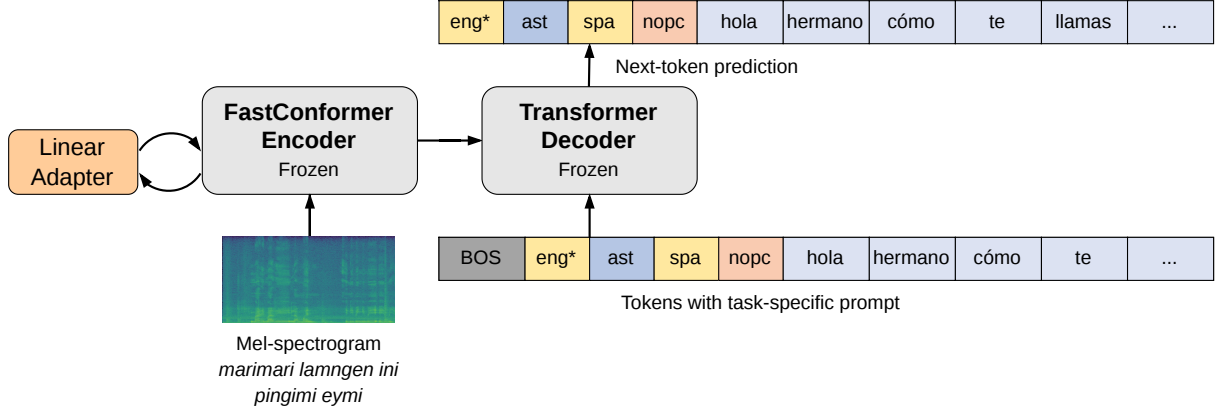


Figure 1: Modified Canary model for AST adopting a PEFT strategy. A Linear Adapter is integrated into the frozen backbone to enable Mapudungun translation. The asterisk in *eng** denotes the use of English as a placeholder for Mapudungun as the source language.

Type	Linear
Input dimension	Model dimension
Output dimension	8
Activation	Swish (default)
Norm position	Pre (default)
Dropout	0.0 (default)

Table 1: Configuration of the adapter

on a linear adapter in the encoder, as shown in Figure 1, which substantially reduces the number of trainable parameters (from 1.7 billion to 589k). Table 1 summarizes the main adapter configuration. We set the translation direction to *eng* $\rightarrow$ *spa* even though our actual source language is Mapudungun (*arn* $\rightarrow$ *spa*). This strategy effectively hijacks one of the predefined languages by using English as a placeholder for Mapudungun. We chose this setting for technical reasons. First, adding a new language to the model would require fine-tuning the entire model. Second, we hypothesized that the speech signal would have a stronger effect on the task because the model already knows how to generate Spanish. In this sense, our main hypothesis is that we can transfer the model’s knowledge of English acoustic patterns to Mapudungun while preserving its ability to translate into Spanish.

Training was conducted using bfloat16 mixed precision and dynamic duration-based audio bucketing via Lhotse (Želasko et al., 2023) to efficiently process variable-length utterances.

3.3 Submitted runs

We developed three system runs:

- **Augmented (Main):** Incorporates transcrip-

tion pre-processing and on-the-fly speech data augmentation. The model was trained on a single GPU for 19 epochs, utilizing early stopping.

- **Vanilla:** Achieved the best performance during our validation and internal testing phases. It shares the training configuration of the augmented run (19 epochs, single GPU, early stopping) and includes transcription pre-processing, but strictly omits speech data augmentation.
- **Filtered:** Follows the vanilla configuration, with an additional data curation step that discards audio recordings exceeding 15 seconds in duration. This model was trained across two GPUs.

Additionally, after submission, we prepared a second version of the third run, which we call **Filtered***. We did this since we had lost a development evaluation during experimentation.

4 Results

The training Word Error Rate (WER) curves show a progressive decrease as expected. Interestingly, the augmented model initially outperforms the vanilla variant; however, by the end of training, the vanilla model surpasses the augmented version by 0.1 WER points. This suggests that data augmentation may offer limited benefit, as the source corpus already contains significant ambient noise. Both models showed a steady decline in loss and were terminated early per our early stopping policy. Final performance metrics are summarized in Table 2.

Run	WER on dev	BLUE on dev
Main	2.5	1.015
Vanilla	2.3	1.05
Filtered*	1.56	3.7

Table 2: These are the Word Error Rate reported for our runs.

Qualitative analysis indicates that our models capture Mapudungun speech phenomena to some extent, as shown in Example 1. Notably, the model successfully transcribes terms like *mapuche* and *huinca*, which are likely underrepresented in the pre-training corpus due to their specific regional context and the model’s high-resource language bias. Conversely, as seen in Example 2, the model tends to fall back on repetitive, basic Spanish structures. This behavior results in higher WER and lower BLEU scores, likely due to the model’s bias toward the higher-resource language.

1. Accurate Rare-Word Transcription

Reference: sí pero yo creo que hay
eso sí que no sé pero en las cosas del
huinca parece que hay una tableta
Predicted: sí pero yo creo que hay
eso sí que no sé pero en el huinca
parece que hay una tableta

2. Pathological Decoding (Repetition)

Reference: en el monte en la que-
brada esta en la montaña
Predicted: en la tierra de la tierra
de la tierra

Overall, the loss, WER, and BLEU learning curves exhibited standard, stable convergence. The fact that training was terminated early suggests that our current architecture, even with modest hardware, has not yet saturated its learning capacity. These observations indicate that performance could be further improved by scaling the data, fine-tuning training configurations, and unfreezing more adapter layers. Beyond configuration tuning, adopting a model that incorporates speech-input pre-training on low-resource languages would allow the architecture to better adapt to the phonetic nuances of the target language.

5 Conclusions

We presented our approach to adapting a multilingual-multitask foundation model for

speech-to-text translation from Mapudungun to Spanish. Despite resource constraints, our methodology successfully captured distinct Mapudungun phonetic features, demonstrating the viability of parameter-efficient fine-tuning for this task.

For future work, we intend to extend this framework to develop broader multilingual translation systems that include other Latin American indigenous languages, such as Quechua and Nahuatl. Ultimately, our research aims to help close the technological divide between high-resource and low-resource languages, ensuring these voices are represented in modern speech technologies—one language at a time.

6 Limitations

In developing our approach, we encountered the following notable constraints:

1. Given the available time and computational resources, we were not able to fine-tune the entire model. We addressed this challenge by using a PEFT training strategy.
2. The chosen model supports English-centric speech translation, which limited our ability to model the direction directly (*arn* → *spa*) and forced us to use a placeholder language. Because we wanted to preserve the model’s knowledge of generating Spanish, we had to rely on a strong English acoustic model.
3. We encountered training-stability issues, particularly when using two GPUs to accelerate training.

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